SETS AND NUMBERS

1. Sets

In math, we view everything as `objects` and mathematics is about *how* these `objects` interact with each other.

Number sets are a collection of `objects` where each object is a number. Some number sets:

$\mathbb{N} = \{0, 1, 2, 3, \ldots\}$	**N**atural numbers (positive whole numbers and 0)
$\mathbb{Z} = \{\ldots, -3, -2, -1, 0, 1, 2, 3, 4 \ldots\}$	**Z**ahlen = integers (non-decimals)
$\mathbb{Q} = \{\text{fractions}\} = \left\{\frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0\right\}$	**Q**uotient (all fractions and all decimals that repeat)
$\mathbb{R} = (-\infty, \infty)$	**R**eal numbers (all numbers that are on the number line)

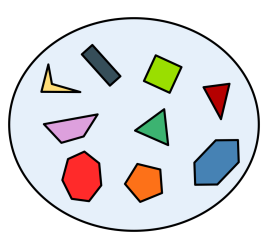
The notation $\in$ is "in". If A is a set, we say that $x \in A$ if x is an `object` that is in A but we use the notation $x \notin A$ if x cannot be found in A.

Example 1.

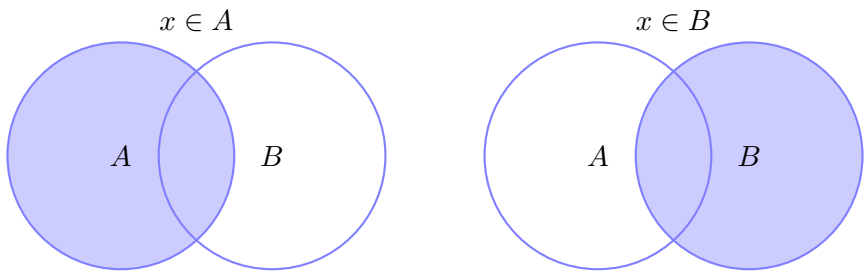
(a) $-3 \in \mathbb{Z}$ but $-3 \notin \mathbb{N}$.
(b) 10th grade $\in$ high school
(c) $\infty \notin \mathbb{R}$ (this says infinity is not in the Real Numbers set)

Since sets are collecion of objects, we like to use venn diagrams to draw sets.

Example 2. Set of polygons with $\triangle \in S$ where S is the set below:

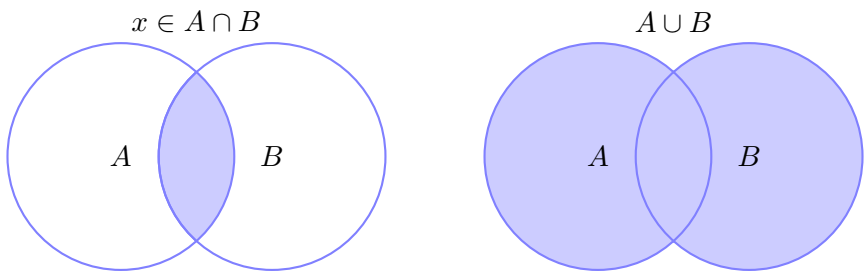
$S =$ 

In general, we may have more than 1 set. Suppose we have two sets A and B. The



We define a few set operations:

- ⋄ the **intersection** of A and B $\quad A \cap B = \{x : x \in A \text{ or } x \in B\}$
- ⋄ the **union** of A and B: $\quad A \cup B = \{x : x \in A \text{ or } x \in B\}$



The intersection only looks at `object` found in both A and B sets

The union looks at `objects` that are found in either A or in B (or both!)

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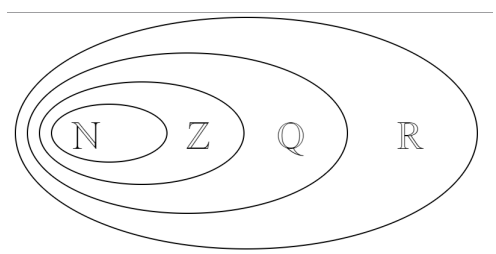
So how does our number sets look like in relation to each other?



2. More number systems

Throughout this year, we will focus most of our studies on $\mathbb{Z}$ and $\mathbb{R}$ but we will also look at other number systems including the two below (no need to know these too well yet)

(1) complex number system

$$\mathbb{C} = \textbf{C}\text{omplex numbers} = \left\{x + yi : x, y \in \mathbb{R}, i^2 = -1\right\}$$

(2) mod n number system (for some positive integer n) which is denoted by $\mathbb{Z}/n\mathbb{Z}$ (or $\mathbb{Z}_n$)

$$\mathbb{Z}/n\mathbb{Z} = \mathbb{Z}_n = \{0, 1, 2, \ldots, n-1\}$$

(a) if $n = 1$, then $\mathbb{Z}/1\mathbb{Z} = \{0\}$
(b) if $n = 2$ then $\mathbb{Z}/2\mathbb{Z} = \{0, 1\}$
(c) if $n = 3$ then $\mathbb{Z}/2\mathbb{Z} = \{0, 1, 2\}$
(d) if $n = 4$, then $\mathbb{Z}/4\mathbb{Z} = \{0, 1, 2, 3\}$
(e) if $n = 12$, then $\mathbb{Z}/12\mathbb{Z} = \{0, 1, 2, 3, \ldots, 10, 11\}$

3. Intervals

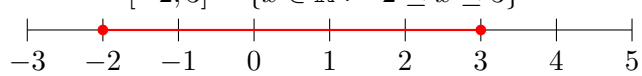
Assuming we know how to compare any two objects in $\mathbb{R}$ (or $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ or irrationals) we can place all of the objects on a number line.

An ***interval*** is a set of the form:

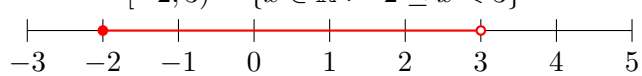
$$[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}$$
$$[a, b) = \{x \in \mathbb{R} : a \leq x < b\}$$
$$(a, b] = \{x \in \mathbb{R} : a < x \leq b\}$$
$$(a, b) = \{x \in \mathbb{R} : a < x < b\}$$

For example we let $a = -2$ and $b = 3$. (Note the circles at the endpoints; filled = included, unfilled = not included)

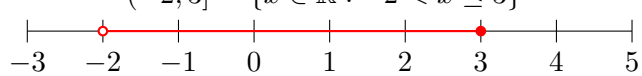
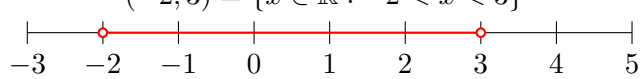
$$[-2, 3] = \{x \in \mathbb{R} : -2 \leq x \leq 3\}$$



$$[-2, 3) = \{x \in \mathbb{R} : -2 \leq x < 3\}$$



$$(-2, 3] = \{x \in \mathbb{R} : -2 < x \leq 3\}$$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